Midterm Review Sheet: Chapters 1 to 3

Exam logistics

- **Date: Thursday, October 22, in class**
- **Time:** full 75 minutes of class period
- **Scope:** K&R Chapters 1 through 3, up to and including TCP flow control and congestion control (3.5 through 3.7)
  - Not included: TCP CUBIC, TCP fairness, and Sections 3.8 and 3.9
- **Format:** expect a mix similar to Quiz 1 and course discussions, including
  - short true/false or conceptual questions;
  - one delay calculation problem;
  - short-answer questions where you explain a mechanism in your own words;
  - at least one timing diagram showing an exchange between a client and a server where you must explain what is happening (e.g., see Problem 13);
  - an open-ended question where you reason about an Internet policy issue similar to our class discussions

Topic checklist

Chapter 1: Computer Networks and the Internet

- Network edge vs. network core; hosts, packet switches, links, access networks (1.1, 1.2)
- Packet switching and store-and-forward; the "network of networks" ISP structure (1.3)
  - Note: circuit switching is not on the exam
- The four components of nodal delay: processing, queuing, transmission, propagation (1.4.1)
- Queuing delay and packet loss; why traffic intensity matters (1.4.2)
- End-to-end delay over multiple links; throughput and the bottleneck link (1.4.3, 1.4.4)
- The five-layer Internet stack and encapsulation (1.5)

Chapter 2: Application Layer

- Client-server vs. P2P; processes, sockets, ports; what transport services apps need (2.1)
- HTTP: persistent vs. non-persistent connections, counting RTTs, cookies, web caching (2.2)
- Email: SMTP, mail servers, what `Received:` headers show (2.3)
- DNS: hierarchy, local resolvers, iterative vs. recursive queries, caching and TTLs, record types (2.4)
- Video streaming and CDNs (2.6)
- Sockets as a concept: the "door" between an application and the transport layer (2.7)

- Know the big picture of what a TCP server and client each do (think back to our in-class acting activity and Lab 1), and why UDP skips the connection setup
- You do not need to memorize socket API calls

Chapter 3: Transport Layer

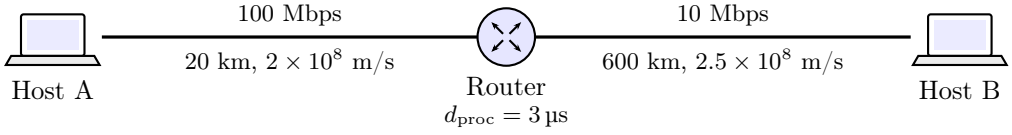
- Transport services; multiplexing and demultiplexing (3.1, 3.2)
- UDP and the checksum (3.3)
- Reliable data transfer: rdt 2.0 to 3.0; stop-and-wait vs. pipelining (3.4.1, 3.4.2)
- Go-Back-N and Selective Repeat (3.4.3, 3.4.4)
  - Will review these concepts during the Tuesday (10/13) worksheet activity
- TCP sequence and ACK numbers, cumulative ACKs, fast retransmit (3.5.2, 3.5.4)
- TCP timeout: EstimatedRTT, DevRTT, TimeoutInterval (3.5.3)
  - Will review these concepts during the Tuesday (10/13) worksheet activity
- Flow control and the three-way handshake (3.5.5, 3.5.6)
- Causes of congestion (the knee and the cliff); end-to-end vs. network-assisted congestion control (3.6)
- TCP congestion control: slow start, AIMD, ssthresh, Reno vs. Tahoe (3.7)

Delay practice problems

Assumptions for all five problems: every packet switch is store-and-forward and its queues start empty. Packets leave the source back-to-back, with no gap between them. "Total time" means from the moment the source starts sending the first bit until the last bit of the last packet arrives at the destination. Ignore processing delay unless it is given. 1 byte = 8 bits.

Problem 1

Host A sends one 1,500-byte packet to Host B through one router. The router has a processing delay of 3 µs.

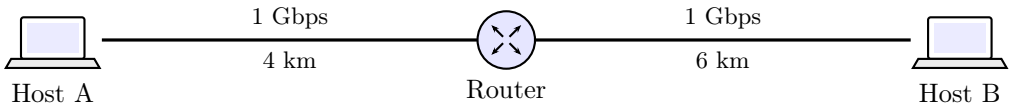


A. Compute each delay component

B. What is the total time? Which single component dominates, and why?

Problem 2

Host A sends three 1,000-byte packets to Host B through one router. Both links have propagation speed 2×10^8 m/s.



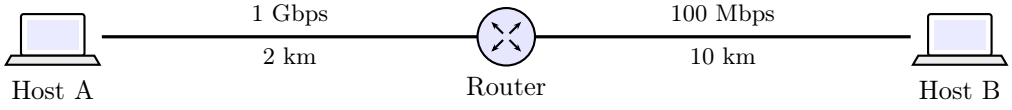
A. Does any packet wait in the router's queue? Explain.

B. What is the total time?

C. Write a general expression for N packets of L bits over two links of rate R

Problem 3

Host A sends four 1,000-byte packets to Host B through one router. Both links have propagation speed 2×10^8 m/s.



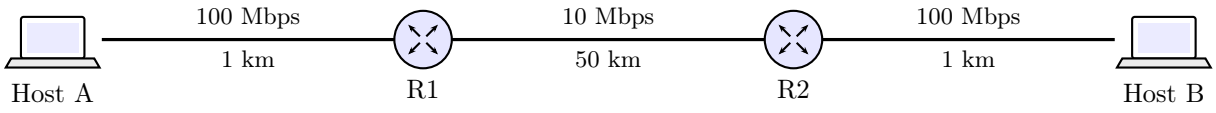
A. When does each packet finish arriving at the router?

B. How long does each packet wait in the router's queue?

C. What is the total time?

Problem 4

Host A sends three 1,250-byte packets to Host B through two routers (R1, then R2). All links have propagation speed 2×10^8 m/s.



A. At which router does queuing happen? Why not at the other one?

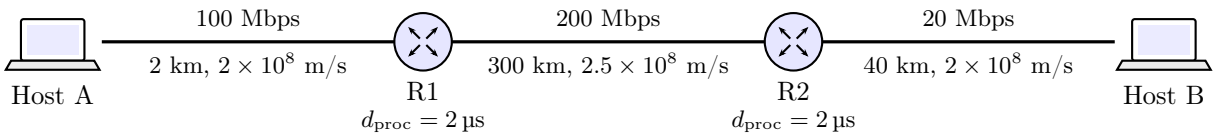
B. What is the queuing delay of each packet?

C. What is the total time?

Problem 5

Optional for extra practice. Actual exam problem will be shorter.

Host A sends five 1,500-byte packets to Host B through two routers. Each router has a processing delay of 2 µs.



A. Where does queuing happen, and how much does each packet wait?

B. What is the total time for all five packets?

C. Now suppose Host A sends the same 7,500 bytes as one single packet instead of five. What is the total time? Explain why breaking a message into packets helps.

D. What is the throughput of this path, and which link sets it?

Concept practice problems

Problem 6: Encapsulation *(1.5)*

You load a web page.

A. Name the unit of data at each layer (application, transport, network, link)

B. Which layers does a router process? Which does a link-layer switch process? Why does the difference matter?

Problem 7: Counting RTTs *(2.2)*

A web page is one base HTML file plus 6 small images, all on the same server. The DNS lookup for the server takes 60 ms in total. The RTT to the server is 20 ms, and transmission times are negligible. How long until the whole page arrives with:

A. non-persistent HTTP, no parallel connections?

B. non-persistent HTTP with up to 6 parallel connections?

C. persistent HTTP without pipelining?

D. persistent HTTP with pipelining?

Problem 8: Tracing a DNS lookup *(2.4)*

A student on campus visits `www.example.org` for the first time, and the local DNS server's cache is empty.

A. List, in order, every DNS server contacted. Assume the host sends a recursive query to its local DNS server, and the local DNS server then uses iterative queries.

B. How would the messages change if every query in the chain were recursive? Which servers would have to do more work?

C. A classmate visits the same site 10 seconds later. What happens now?

D. Someone then emails `you@example.org`. Which record type is needed, and which servers can be skipped this time?

Problem 9: Sockets *(2.7 and 3.2)*

A. A TCP server (like your Lab 1) has 3 clients connected. How many sockets does the server have open, and why?

B. A UDP-based server receives messages from the same 3 clients. How many sockets does it need? Explain using demultiplexing.

Problem 10: Demultiplexing *(3.2)*

Host C runs a web server on port 80. Host A opens two TCP connections to it, from ports 5000 and 5001. Host B opens one, from port 5000.

A. Write the source and destination ports for a segment from each connection, in both directions

B. How does C tell A's port 5000 apart from B's port 5000?

Problem 11: Building rdt *(3.4.1)*

Each scenario below is a problem that can happen when sending data over an unreliable channel. For each one, name the mechanism that lets the sender and receiver handle it, and the earliest rdt version (2.0, 2.1, 2.2, or 3.0) that includes that mechanism.

A. Some bits in a data packet get flipped on the way to the receiver

B. The ACK or NAK coming back to the sender is corrupted, so the sender cannot tell which one it was

C. The receiver gets a second copy of a packet it has already delivered

D. A data packet never arrives at the receiver

Problem 12: Go-Back-N *(3.4.3)*

A GBN sender has window size $N = 4$ and sends packets 0, 1, 2, 3. Packet 2 is lost; everything else arrives, and ACKs arrive before any timeout.

A. What ACKs does the receiver send?

B. Which new packets does the sender send as the window slides?

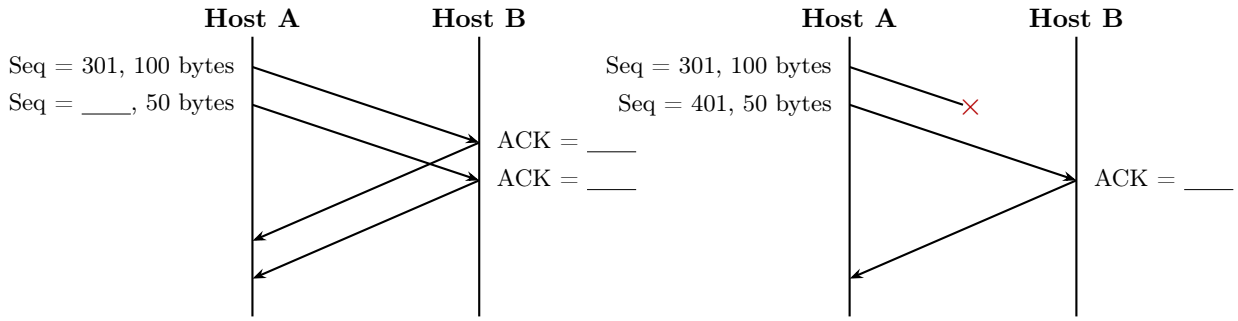
C. When packet 2's timer expires, what gets resent?

TCP problems

- Assume the receiver ACKs every segment
- "3 duplicate ACKs" means 4 ACKs with the same number: the original plus 3 repeats
- cwnd changes once per transmission round
  - In slow start it doubles each round until it reaches ssthresh (if doubling would pass ssthresh, cwnd is set to ssthresh)
  - After that it grows by 1 MSS per round
- On 3 duplicate ACKs: ssthresh = cwnd/2
  - Reno sets cwnd = ssthresh (the textbook's fast recovery adds 3 MSS to Reno's value; you may ignore that detail)
  - Tahoe sets cwnd = 1 MSS
- On a timeout: ssthresh = cwnd/2 and cwnd = 1 MSS, for both Reno and Tahoe
- The sender can have at most min(cwnd, rwnd) unacknowledged bytes in flight: LastByteSent − LastByteAcked ≤ min(cwnd, rwnd)

Problem 13: TCP sequence and ACK numbers *(3.5.2, 3.5.4)*

Host B has received every byte from Host A up through byte 300. A then sends two segments back-to-back: 100 bytes, then 50 bytes. The diagrams below show two versions of what happens next.



(1) Both segments arrive (2) First segment lost

A. In diagram (1), fill in the missing sequence number and both ACK numbers

B. In diagram (2), what ACK number does B send? Why that number?

C. What event would make Host A retransmit before its timer expires? Would that event happen here? Why or why not?

Problem 14: Flow control *(3.5.5)*

B's receive buffer is 4,096 bytes. B has received 2,560 bytes the application has not read yet.

A. What rwnd does B advertise?

B. What does A do if rwnd drops to 0?

Problem 15: Congestion control *(3.7)*

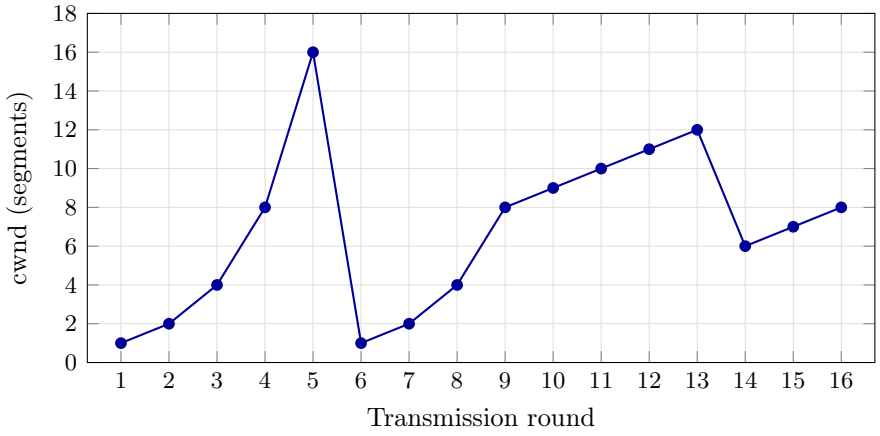
A TCP sender starts with cwnd = 1 MSS and ssthresh = 16 MSS.

A. What is cwnd in rounds 1 through 9? In which round does slow start end?

B. After round 9, the sender gets 3 duplicate ACKs. What are the new ssthresh and cwnd under TCP Reno? Under TCP Tahoe?

C. Later, a timeout happens when cwnd = 12. What are the new ssthresh and cwnd?

Problem 16: Congestion control graph *(3.7)*
The graph below shows the congestion window of a TCP sender over 16 transmission rounds.



A. Identify the rounds where TCP is in slow start and the rounds where it is in congestion avoidance

B. What is ssthresh during rounds 6 to 13? During rounds 14 to 16?

C. Explain what happens between rounds 13 and 14. Is this sender running TCP Reno or TCP Tahoe? How can you tell?

D. Where does a timeout happen? How can you tell from the graph?

E. What can you say about the value of ssthresh at the start (round 1)?

Problem 17: Duplicate ACKs *(3.5.4)*

Host A sends seven 100-byte segments with sequence numbers 1, 101, 201, 301, 401, 501, 601. The segment with sequence number 201 is lost; all others arrive.

A. List the ACKs B sends, in order

B. After which ACK does A fast retransmit, and which segment does it resend?

C. When the resent segment arrives, what ACK does B send?

Problem 18: Flow and congestion control together *(3.5.5, 3.7)*

Host A is sending a large file to Host B over TCP, with an MSS of 1,000 bytes.

A. A's cwnd is 10 MSS, and B advertises rwnd = 6,000 bytes. How many bytes can A have sent but not yet acknowledged? Is flow control or congestion control limiting A?

B. Later, A's cwnd is 4 MSS, and B advertises rwnd = 20,000 bytes. Answer the same two questions.

C. In the situation from part B, the RTT is 50 ms. Roughly how fast is A sending, in bits per second?

D. Still in the situation from part B, A has sent up through byte 9,000, and B has acknowledged up through byte 6,000. How many more bytes can A send before it has to wait for an ACK?

Problem 19: Short answers *(3.5, 3.6, 3.7)*

A. Flow control and congestion control both slow the sender down. Why doesn't rwnd prevent congestion?

B. Why does TCP Reno treat 3 duplicate ACKs as less serious than a timeout?

C. Why does the handshake need three messages instead of two?

D. On the knee and cliff graph, what happens to goodput and delay as load passes the knee, and then the cliff?

For practice reading a cwnd graph, try textbook P40 and the "TCP congestion window evolution" problems on the K&R interactive site (linked below).

More practice

You may also take a look at end-of-chapter problems in the textbook as a review.

Optional: Hands-on DNS review

Spend about 15 minutes on Mess With DNS, which gives you a real subdomain you can create records for and watch queries arrive. Do the **3 tutorial experiments** and the **first 4 "weird" experiments**.

As you go, connect each experiment back to Section 2.4:

- When a request shows up in the log, who sent it: your browser, your local resolver, or something else? Is that an iterative or recursive query?
- What role does the TTL play? What happens when you change a record and query again right away?

Other interesting resources

- K&R interactive problems: free, auto-generated versions of textbook exercises with answers, so you can practice. Most useful for the midterm: end-to-end delay, queuing delay, throughput, DNS, TCP sequence and ACK numbers, and TCP congestion window evolution.
- Cloudflare: Benefits of a CDN: a short summary of why CDNs exist. Read it alongside Section 2.6 and connect each benefit to delay, throughput, or server load.
- Cloudflare: What is DNS over TLS?: an applied example that ties together several course ideas. Think about how this connects to what you learned about ports, UDP vs. TCP, and who can see your DNS queries.
- ICANN: Who Runs the Internet?: a map of the many organizations involved in running the Internet, a good companion to our governance discussions